WORKED SOLUTION

STEP III 2014, Question 9

Mechanics • Projectiles with Resistance

STEP III 2014, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q9

The key idea

We're given the position, so the velocity is one differentiation away, and checking the equation of motion is just a matter of putting $\mathbf{v}$ and $\dot{\mathbf{v}}$ into $\dot{\mathbf{v}} = \mathbf{g} - k\mathbf{v}$. After that, everything is about components: the $\mathbf{j}$-component of $\mathbf{r}$ fixes T, and the ratio of the components of $\mathbf{v}$ at time T gives $\tan\beta$. The final inequality looks messy until we write $x = kT$ and clear the denominators. Then it collapses to exactly the hint $\sinh x > x$.

Solution

The velocity, and checking the motion

We model the body as a particle. Newton's second law gives $m\dot{\mathbf{v}} = m\mathbf{g} - mk\mathbf{v}$, so the equation of motion is

$$\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{g} - k\mathbf{v},$$

and the initial conditions are $\mathbf{r} = \mathbf{0}$ and $\mathbf{v} = \mathbf{u}$ at $t = 0$.

Here $\mathbf{g}$ and $\mathbf{u}$ are constant vectors, so we differentiate the given $\mathbf{r}$ coefficient by coefficient:

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{k - k\mathrm{e}^{-kt}}{k^2}\,\mathbf{g} + \frac{k\mathrm{e}^{-kt}}{k}\,\mathbf{u} = \frac{1 - \mathrm{e}^{-kt}}{k}\,\mathbf{g} + \mathrm{e}^{-kt}\,\mathbf{u}.$$

Differentiating again gives $\dot{\mathbf{v}} = \mathrm{e}^{-kt}\mathbf{g} - k\mathrm{e}^{-kt}\mathbf{u}$. On the other hand,

$$\mathbf{g} - k\mathbf{v} = \mathbf{g} - (1 - \mathrm{e}^{-kt})\mathbf{g} - k\mathrm{e}^{-kt}\mathbf{u} = \mathrm{e}^{-kt}\mathbf{g} - k\mathrm{e}^{-kt}\mathbf{u},$$

which is the same, so the equation of motion holds. At $t = 0$ we have $\mathrm{e}^0 = 1$, so $\mathbf{r} = \frac{0-1+1}{k^2}\mathbf{g} + 0\,\mathbf{u} = \mathbf{0}$ and $\mathbf{v} = 0\,\mathbf{g} + \mathbf{u} = \mathbf{u}$. Both initial conditions hold.

The time of flight

Now take $\mathbf{u} = u\cos\alpha\,\mathbf{i} + u\sin\alpha\,\mathbf{j}$ and $\mathbf{g} = -g\,\mathbf{j}$. The $\mathbf{j}$-component of $\mathbf{r}$ is

$$\mathbf{r}\cdot\mathbf{j} = -\frac{kt - 1 + \mathrm{e}^{-kt}}{k^2}\,g + \frac{1-\mathrm{e}^{-kt}}{k}\,u\sin\alpha.$$

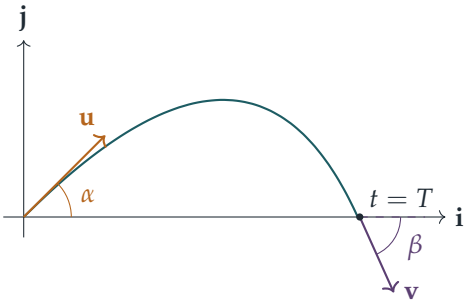
Setting this to zero at $t = T$ and multiplying by k^2:

$$uk\sin\alpha\,(1-\mathrm{e}^{-kT}) = (kT - 1 + \mathrm{e}^{-kT})g.$$

Since $T > 0$ and $k > 0$, we have $1 - \mathrm{e}^{-kT} > 0$, so we may divide by it. Writing $kT - 1 + \mathrm{e}^{-kT} = kT - (1 - \mathrm{e}^{-kT})$,

$$uk\sin\alpha = \left(\frac{kT}{1-\mathrm{e}^{-kT}} - 1\right)g,$$

as required.



The angle at time T

At $t = T$ the components of $\mathbf{v} = \frac{1-\mathrm{e}^{-kT}}{k}\mathbf{g} + \mathrm{e}^{-kT}\mathbf{u}$ are

$$v_x = \mathrm{e}^{-kT}u\cos\alpha, \qquad v_y = -\frac{1-\mathrm{e}^{-kT}}{k}\,g + \mathrm{e}^{-kT}u\sin\alpha.$$

Since $0 < \alpha < 90°$, $v_x > 0$. We expect $v_y < 0$, since the particle is coming back down, and we'll prove it below. So the acute angle between $\mathbf{v}$ and $\mathbf{i}$ satisfies $\tan\beta = -v_y/v_x$. Dividing through by v_x:

$$\tan\beta = \frac{(1-\mathrm{e}^{-kT})g}{k\,\mathrm{e}^{-kT}u\cos\alpha} - \frac{\mathrm{e}^{-kT}u\sin\alpha}{\mathrm{e}^{-kT}u\cos\alpha} = \frac{(\mathrm{e}^{kT}-1)g}{uk\cos\alpha} - \tan\alpha,$$

using $(1-\mathrm{e}^{-kT})/\mathrm{e}^{-kT} = \mathrm{e}^{kT} - 1$.

Showing $\tan\beta > \tan\alpha$

From the last result, $\tan\beta > \tan\alpha$ is the same as

$$\frac{(\mathrm{e}^{kT}-1)g}{uk\cos\alpha} > 2\tan\alpha \iff (\mathrm{e}^{kT}-1)g > 2uk\sin\alpha,$$

multiplying by $uk\cos\alpha > 0$. Now we use the time-of-flight result to replace $uk\sin\alpha$, which removes u and α altogether.

Ask yourself How do we turn this into something involving $\sinh kT$?

Write $x = kT > 0$. Then $\frac{x}{1-e^{-x}} = \frac{xe^x}{e^x-1}$, and after dividing by $g > 0$ the inequality we need becomes

$$e^x - 1 > \frac{2xe^x}{e^x - 1} - 2.$$

Multiply by $e^x - 1 > 0$ (so the direction is kept) and expand:

$$\begin{aligned}
&\iff e^{2x} - 2e^x + 1 > 2xe^x - 2e^x + 2 \\
&\iff e^{2x} - 1 > 2xe^x \\
&\iff e^x - e^{-x} > 2x \qquad \text{(dividing by } e^x > 0\text{)} \\
&\iff \sinh x > x.
\end{aligned}$$

Every step is reversible, and $\sinh kT > kT$ for $kT > 0$ (we're allowed to assume this). So $\tan\beta > \tan\alpha$.

This also settles the sign of v_y: we've shown $-v_y/v_x > \tan\alpha > 0$ and $v_x > 0$, so $v_y < 0$. The particle really is descending at time T, and the formula for $\tan\beta$ is justified.

Finally, α and β both lie in $(0°, 90°)$, where tan is strictly increasing. So $\tan\beta > \tan\alpha$ gives $\beta > \alpha$. With resistance, the particle comes down more steeply than it went up.

Checks

- As $k \to 0$, $\mathbf{r} \to \frac{1}{2}t^2\mathbf{g} + t\mathbf{u}$, using $e^{-kt} \approx 1 - kt + \frac{1}{2}k^2t^2$. That's the usual projectile. Also $\frac{x}{1-e^{-x}} - 1 \approx \frac{x}{2}$ for small x, so the time of flight gives $T \to 2u\sin\alpha/g$. Similarly $\tan\beta \to \frac{gT}{u\cos\alpha} - \tan\alpha = \tan\alpha$, the symmetric parabola.
- Take $k = 1\,\mathrm{s}^{-1}$, $g = 10\,\mathrm{m\,s}^{-2}$, $u = 10\,\mathrm{m\,s}^{-1}$ and $\alpha = 45°$. Solving $\mathbf{r}\cdot\mathbf{j} = 0$ numerically gives $T \approx 1.185\,\mathrm{s}$, shorter than the $1.414\,\mathrm{s}$ with no resistance. The landing angle is $\beta \approx 65.7° > 45°$, and both formulae check out to many decimal places. A step-by-step numerical integration of $\dot{\mathbf{v}} = \mathbf{g} - k\mathbf{v}$ by computer reproduces the given $\mathbf{r}$ and $\mathbf{v}$.

Common slips

- Taking $\tan\beta = v_y/v_x$ without the minus sign: v_y is negative at landing, and β is acute.
- Dividing or multiplying by $1 - e^{-kT}$ or $e^{kT} - 1$ without saying that it's positive. That's what keeps the inequality the right way round.
- Stopping at $\tan\beta > \tan\alpha$: the deduction $\beta > \alpha$ needs the fact that tan is increasing on the acute angles.

Beyond STEP

The given $\mathbf{r}$ solves the linear differential equation $\ddot{\mathbf{r}} + k\dot{\mathbf{r}} = \mathbf{g}$, which you can solve directly with an integrating factor e^{kt}. Linear drag is the one resistance law where the $\mathbf{i}$- and $\mathbf{j}$-motions separate. With the more realistic quadratic drag $-mk|\mathbf{v}|\mathbf{v}$ they become coupled, and no closed form exists. The lopsided trajectory, steeper on the way down, holds in both cases. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion with resistive forces in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.